

A proposed proof of Erdős Problem 689

Working note

April 25, 2026

Abstract

This note records a proposed proof of Erdős Problem 689 for all sufficiently large n . The proof uses a fixed small-prime sieve, robust cleanup primes larger than $n/5$, a finite-core prime-difference matching, the Green–Tao finite-complexity linear-forms theorem in primes, made unconditional by the Green–Tao nilsequence theorem and the Green–Tao–Ziegler inverse theorem, and Kahn’s fractional Frankl–Rödl–Pippenger rounding theorem. The note is written as a proof sketch with enough detail to expose the interfaces between the elementary construction, the GTZ input, and the hypergraph matching input.

Public metadata and previews for Kahn’s paper appear to state the relevant pair co-load parameter and the needed fractional-rounding conclusion. A final public version should still verify the printed Theorem 1.5 directly. This note was prepared with AI assistance and should be independently checked.

Relation to the existing EP689 discussion. The discussion thread for EP689 already points toward this technology: earlier comments suggest using linear equations in primes to control averaged prime-pattern counts and Pippenger–Spencer/Kahn-style hypergraph covering to turn fractional coverage into a genuine disjoint cover. The new feature of the present note is the robust $P > n/5$ cleanup setup and the finite-core prime-difference matching formulation.

1 Statement and conventions

For a prime $p \leq n$, write $a_p \pmod{p}$ for the residue class chosen for p . The goal is to choose the a_p ’s so that every integer $m \in [1, n]$ is hit by at least two chosen congruence classes:

$$\#\{p \leq n : m \equiv a_p \pmod{p}\} \geq 2.$$

We leave most primes in their zero class. The prime 2 is put in the odd class $a_2 \equiv 1 \pmod{2}$, and 3 is left at 0 $\pmod{3}$. A fixed finite set $S \subset \{7, 11, 13, \dots\}$ is switched to nonzero residues $b_s \pmod{s}$. Further primes $P > n/5$ will be used as cleanup primes.

For the fixed set S , put

$$H_S(x) = \#\{s \in S : x \equiv b_s \pmod{s}\}.$$

After the fixed switches in S , the main residual demand consists of even integers of the form

$$x = 2^k u q, \quad k \geq 1,$$

where u is odd S -smooth and $q \notin S$ is an odd prime appearing to exponent one, subject to the finite system of congruence exclusions $H_S(x) = 0$. Prime powers, pure $\{2\} \cup S$ -smooth terms, and

small-prime boundary cases contribute only $o(n/\log n)$ residual tokens. Standard fixed-modulus prime number theory and the identity

$$\sum_{k \geq 1} 2^{-k} \sum_{A \subseteq S} \prod_{s \in A} \frac{1}{s-1} \prod_{s \notin A} \left(1 - \frac{1}{s-1}\right) = 1$$

give

$$|A_S(n)| = (1 + o(1)) \frac{n}{\log n}.$$

Moreover, if

$$A_1(n) = \{x \in A_S(n) : v_2(x) = 1\}, \quad A_2(n) = \{x \in A_S(n) : v_2(x) \geq 2\},$$

then

$$|A_1(n)| = \left(\frac{1}{2} + o(1)\right) |A_S(n)|, \quad |A_2(n)| = \left(\frac{1}{2} + o(1)\right) |A_S(n)|.$$

2 Robust cleanup primes

Let

$$W = \prod_{s \in S} s.$$

For a unit residue $\pi \pmod{W}$, call π robust if

$$H_S(\pi) \geq 1, \quad H_S(2\pi) \geq 2, \quad H_S(4\pi) \geq 2.$$

Let $\mathcal{B} \subset (\mathbb{Z}/W\mathbb{Z})^\times$ be the set of robust classes and

$$\delta_S = \frac{|\mathcal{B}|}{\varphi(W)}.$$

For each $s \in S$, the three classes

$$\pi \equiv b_s, \quad 2\pi \equiv b_s, \quad 4\pi \equiv b_s \pmod{s}$$

are distinct. Thus δ_S depends only on S , not on the particular nonzero residues b_s . A union bound gives

$$\delta_S \geq 1 - A_S^{(0)}(3 + 2B_S),$$

where

$$A_S^{(0)} = \prod_{s \in S} \left(1 - \frac{1}{s-1}\right), \quad B_S = \sum_{s \in S} \frac{1}{s-2}.$$

For instance, take $S(y) = \{p : 7 \leq p \leq y, p \text{ prime}\}$. By Mertens' theorem and the prime number theorem,

$$A_{S(y)}^{(0)} \asymp \frac{1}{\log y}, \quad B_{S(y)} = O(\log \log y),$$

so $A_{S(y)}^{(0)}(3 + 2B_{S(y)}) \rightarrow 0$. Hence we may choose a fixed finite S so that δ_S is as close to 1 as needed. This S may be very large; its size is fixed before the limit $n \rightarrow \infty$.

Set

$$\beta_* = \frac{1}{2} \left(1 - \frac{3}{5}e^{-2}\right) = 0.459399 \dots$$

and

$$\delta_* = \frac{1}{\beta_* + 3/5} = 0.94393 \dots$$

Choose S so that $\delta_S > \delta_*$, and then choose

$$\delta_S^{-1} - \frac{3}{5} < \beta < \beta_*. \quad (1)$$

Let

$$\Delta = (\beta + \frac{3}{5})\delta_S - 1 > 0. \quad (2)$$

This fixed positive margin is the surplus that later absorbs coefficient tails and the $o(n/\log n)$ exceptional residual tokens. Let

$$\mathcal{R}_\beta(n) = \{P \in (n/5, \beta n] : P \text{ prime, } P \bmod W \in \mathcal{B}\}.$$

By the prime number theorem in fixed arithmetic progressions,

$$|\mathcal{R}_\beta(n)| = ((\beta - \frac{1}{5})\delta_S + o(1)) \frac{n}{\log n}.$$

The full robust reservoir

$$\mathcal{R}(n) = \{P \in (n/5, n] : P \text{ robust}\}$$

has size

$$|\mathcal{R}(n)| = \left(\frac{4}{5}\delta_S + o(1)\right) \frac{n}{\log n}.$$

Lemma 1 (Robust side-debt). *Switching a robust prime $P > n/5$ to any nonzero residue creates no unresolved side debt.*

Proof. The only multiples of $P \leq n$ are $P, 2P, 3P, 4P$. The integer P is covered by the parity class and at least one S -hit, since $H_S(P) \geq 1$. The integers $2P$ and $4P$ have at least two S -hits by robustness. The integer $3P$ is covered by the parity class and the unchanged zero class modulo 3. There is no multiple $5P \leq n$. $\square$

Consequently, switching any collection of robust primes $P > n/5$ to nonzero residues creates no new residual tokens: the only zero-class hit lost at a multiple of such a P is replaced by two hits not using the zero class of that P .

3 The matching hypergraph

Choose finite coefficient cores

$$\mathcal{A}_1 \subset \{a : a \text{ odd } S\text{-smooth}\},$$

$$\mathcal{A}_2 \subset \{b : b = 2^j u, j \geq 1, u \text{ odd } S\text{-smooth}\}.$$

The core $X_n \subset A_1(n)$ consists of residual targets $x = 2aq$ with $a \in \mathcal{A}_1$. The core $Y_n \subset A_2(n)$ consists of residual targets $y = 2bq'$ with $b \in \mathcal{A}_2$. The finite cores are chosen to capture fractions α_X, α_Y of the A_1, A_2 coefficient mass, uniformly in the half-residue classes defined below, with

$$\alpha_X, \alpha_Y > G(\beta) + \eta$$

for some fixed $\eta > 0$, where

$$G(\beta) = \int_{1/5}^{\beta} \frac{dt}{1-2t} = \frac{1}{2} \log \left(\frac{3/5}{1-2\beta} \right) < 1. \quad (3)$$

This is possible because the full S -smooth coefficient distributions are absolutely summable. The order of quantifiers is: after fixing S , β , and Δ , choose the coefficient cores so that the omitted main residual coefficient share is smaller than a fixed fraction of Δ and so that the above side-load inequalities hold. Then take n large enough that the actual coefficient-tail counts, the exceptional residual-token count, and all GTZ/Kahn losses are smaller than the remaining margin.

Let $Z_n = \mathcal{R}_\beta(n)$. Define a 3-partite 3-uniform hypergraph

$$\mathcal{H}_n \subset X_n \times Y_n \times Z_n$$

by putting an edge (x, y, P) whenever

$$|y - x| = 2P.$$

The parity split is essential: if both targets had the same 2-adic parity class, then $(y - x)/2$ would be even and could not equal the odd prime P .

The goal is to find a matching in $\mathcal{H}_n$ covering

$$(1 - o(1))|Z_n|$$

labels. Such a matching covers that many pairs of residual targets using one robust prime per pair.

4 A finite-core fractional matching

Put

$$c_s \equiv 2^{-1}b_s \pmod{s}$$

and define the set of residual half-residues

$$\mathcal{C} = \{A \bmod W : A \not\equiv c_s \pmod{s} \text{ for every } s \in S\}.$$

For $x = 2aq$, write $A \equiv aq \pmod{W}$; for $y = 2bq'$, write $B \equiv bq' \pmod{W}$. Then residual membership is simply

$$A, B \in \mathcal{C}.$$

For every unit $\pi \pmod{W}$ and every orientation $\sigma = \pm 1$,

$$\#\{A \in \mathcal{C} : A + \sigma\pi \in \mathcal{C}\} = M := \prod_{s \in S} (s - 2). \quad (4)$$

Locally this just excludes two distinct residues modulo each $s \in S$.

In the aggregate continuum model, a label is (t, π) with

$$t = P/n \in (1/5, \beta], \quad \pi \in \mathcal{B}.$$

For each label, choose $\sigma = \pm 1$ with equal probability, choose uniformly an $A \in \mathcal{C}$ with $A + \sigma\pi \in \mathcal{C}$, and choose $u \in (0, 1 - 2t)$ uniformly. For $\sigma = +1$, send mass to

$$X = (u, A), \quad Y = (u + 2t, A + \pi),$$

and for $\sigma = -1$, send mass to

$$X = (u + 2t, A), \quad Y = (u, A - \pi).$$

This gives exact label load $L_Z(t, \pi) = 1$. The side load is bounded by

$$L_X, L_Y \leq \frac{G(\beta)}{\alpha_X} + o(1), \quad \frac{G(\beta)}{\alpha_Y} + o(1),$$

so by the choice of core it is at most $1 - 2\gamma$ for some fixed $\gamma > 0$.

The aggregate transport lifts to typed kernels. A type is

$$\tau = (a, b, \sigma, r, r', \pi),$$

where $a \in \mathcal{A}_1$, $b \in \mathcal{A}_2$,

$$q \equiv r \pmod{W}, \quad q' \equiv r' \pmod{W},$$

and

$$\pi \equiv \sigma(br' - ar) \pmod{W}, \quad \pi \in \mathcal{B}.$$

The support polygon is

$$D_\tau = \left\{ (Q, Q') : 0 < Q \leq \frac{1}{2a}, \ 0 < Q' \leq \frac{1}{2b}, \ \frac{1}{5} < \sigma(bQ' - aQ) \leq \beta \right\}.$$

Proposition 2 (Typed-kernel lift). *The aggregate transport above can be lifted to bounded kernels g_τ on the finitely many admissible typed polygons D_τ . These kernels have limiting label load 1 on Z_n , and limiting side loads at most*

$$G(\beta)/\alpha_X + o_{\text{core}}(1), \quad G(\beta)/\alpha_Y + o_{\text{core}}(1).$$

Proof. For an X -type $\alpha = (a, r)$, put $A(\alpha) \equiv ar \pmod{W}$, and let $m_X(\alpha)$ be its limiting density in the coordinate $z = x/n$. Define $m_X(A) = \sum_{A(\alpha)=A} m_X(\alpha)$, and define $m_Y(B)$ similarly. The infinite S -smooth coefficient distribution is uniform over $\mathcal{C}$: locally at $s \in S$, the mass of the zero half-residue is $\sum_{e \geq 1} s^{-e} = 1/(s-1)$, while each nonzero half-residue has prime-class density $1/(s-1)$. Finite cores approximate this uniformly, so

$$m_X(A) = \alpha_X m_X^0(1 + o_{\text{core}}(1)), \quad m_Y(B) = \alpha_Y m_Y^0(1 + o_{\text{core}}(1))$$

uniformly in $A, B \in \mathcal{C}$.

Disintegrate the finite-core mass by

$$\rho_X^A(\alpha) = m_X(\alpha)/m_X(A), \quad \rho_Y^B(\beta') = m_Y(\beta')/m_Y(B).$$

For an admissible type $\tau = (a, b, \sigma, r, r', \pi)$, let

$$A = ar, \quad B = br' \pmod{W}, \quad B = A + \sigma\pi,$$

and put $t = \sigma(bQ' - aQ)$. Before dividing by the fixed local GTZ density, assign edge intensity proportional to

$$\frac{2ab}{2M(1-2t)} \rho_X^A(a, r) \rho_Y^B(b, r')$$

on D_τ , with the fixed label-density normalization included. The Jacobian identity

$$dz_X dt = 2ab dQ dQ', \quad z_X = 2aQ,$$

and the disintegrations above show that summing over all typed lifts with fixed (t, π) reproduces the aggregate label mass exactly. Summing over the typed lifts incident to a side vertex reproduces the aggregate side load, but with the finite-core base density divided by the captured fractions α_X, α_Y . Finally divide each typed intensity by the positive fixed GTZ local constant for $q, q', \sigma(bq' - aq)$. Since the type set is finite and $t \leq \beta < 1/2$, the resulting g_τ 's are bounded. $\square$

5 GTZ averaged moments

For an edge $e = (x, y, P)$ of type τ , with

$$x = 2aq, \quad y = 2bq', \quad P = \sigma(bq' - aq),$$

assign weight

$$w_e = \frac{\log^2 n}{n} g_\tau(q/n, q'/n).$$

We use the Green–Tao finite-complexity theorem for affine-linear forms in primes, made unconditional using the Green–Tao nilsequence theorem and the Green–Tao–Ziegler inverse theorem for Gowers norms. The fixed modulus $2W$ is part of the residue data. In a fully expanded proof one may either use an auxiliary growing $W_{\text{GTZ}}(n)$ -trick compatible with these fixed residue classes, or work with fixed-modulus singular series and verify the corresponding conditional local-factor disintegrations. The coefficient cores, residue data, and kernels are fixed before $n \rightarrow \infty$.

Proposition 3 (Weighted moment proposition). *The loads*

$$L_Z(P) = \sum_{e \ni P} w_e, \quad L_X(x) = \sum_{e \ni x} w_e, \quad L_Y(y) = \sum_{e \ni y} w_e$$

satisfy

$$\sum_{P \in Z_n} (L_Z(P) - 1)^2 = o(|Z_n|),$$

$$\sum_{x \in X_n} (L_X(x) - L_X^{\text{lim}}(x))^2 = o(|X_n|),$$

and

$$\sum_{y \in Y_n} (L_Y(y) - L_Y^{\text{lim}}(y))^2 = o(|Y_n|),$$

where the limiting side loads are bounded by $1 - 2\gamma$.

Proof sketch. The edge totals use the three linear forms

$$q, \quad q', \quad \sigma(bq' - aq).$$

The label second moment, after parametrizing by (P, q_1, q_2) , uses

$$P, \quad q_1, \quad \frac{a_1 q_1 + \sigma_1 P}{b_1}, \quad q_2, \quad \frac{a_2 q_2 + \sigma_2 P}{b_2}$$

on a fixed affine lattice. The X -side second moment uses

$$q, \quad q'_1, \quad q'_2, \quad \sigma_1(b_1 q'_1 - aq), \quad \sigma_2(b_2 q'_2 - aq),$$

and the Y -side moment is symmetric. After removing repeated-edge diagonals, no two forms in any system are rationally affinely dependent. The diagonal contribution is

$$O\left(\frac{\log^4 n}{n^2} \cdot \frac{n^2}{\log^3 n}\right) = O(\log n) = o(n/\log n).$$

For primes outside the fixed exceptional set dividing $2Wab$, the local systems are admissible because no two surviving forms are rationally affinely dependent; the small primes are handled by the imposed residue classes. Since the type set is finite, the complexity and error terms are uniform over all types. Thus all required first and second moments are finite-complexity GTZ applications. In the auxiliary W_{GTZ} -tricked formulation the second-moment main terms disintegrate into the limiting load integrals; in the fixed singular-series formulation this is the same conditional local-factor identity. No pointwise estimate for fixed $P = bq' - aq$ is used. $\square$

6 Fractional rounding

Normalize the label loads by replacing w_e with $w'_e = c_P w_e$, where

$$c_P = \begin{cases} \min(1, L_Z(P)^{-1}), & L_Z(P) > 0, \\ 1, & L_Z(P) = 0 \end{cases}$$

for the unique label $P \in e$. The L^2 estimate on Z_n gives

$$\sum_e w'_e = \sum_{P \in Z_n} \min(L_Z(P), 1) = |Z_n| - o(|Z_n|).$$

Delete all edges incident to side vertices with normalized side load > 1 . Here is the standard mass-loss estimate. Let

$$B_X = \{x : L'_X(x) > 1\}.$$

Since $L'_X(x) \leq L_X(x)$ and $L_X^{\lim}(x) \leq 1 - 2\gamma$, every $x \in B_X$ satisfies

$$|L_X(x) - L_X^{\lim}(x)| \geq 2\gamma.$$

Thus the L^2 estimate gives $|B_X| = o(|X_n|)$. Moreover

$$\sum_{x \in B_X} L'_X(x) \leq \sum_{x \in B_X} L_X(x) \leq (1 - 2\gamma)|B_X| + |B_X|^{1/2} \left(\sum_x (L_X(x) - L_X^{\lim}(x))^2 \right)^{1/2} = o(|Z_n|).$$

The same argument applies to the Y -side. Hence deleting overloaded side vertices loses only $o(|Z_n|)$ total mass. The remaining weights $t_e \leq w_e$ form a fractional matching:

$$\sum_{e \ni v} t_e \leq 1 \quad \text{for every vertex } v,$$

with

$$\sum_e t_e = (1 - o(1))|Z_n|.$$

Also

$$\max_e t_e \ll \frac{\log^2 n}{n} = o(1).$$

The hypergraph has codegree at most 2: a pair (x, y) determines $P = |y - x|/2$, and a pair (x, P) or (y, P) has at most two extensions. Therefore the fractional pair co-load satisfies

$$a(t) = \max_{u \neq v} \sum_{e \supset \{u, v\}} t_e \leq 2 \max_e t_e = o(1).$$

By Kahn's fractional Frankl–Rödl–Pippenger theorem, applied with the single statistic $C(e) \equiv 1$, this fractional matching rounds to an actual matching M_n . For this statistic, Kahn's quadratic-smallness condition is

$$\sum_e t_e = o \left(\left(\sum_e t_e \right)^2 \right),$$

which holds because $\sum_e t_e = (1 - o(1))|Z_n| \asymp n/\log n$. Subject to checking the printed statement of Kahn's Theorem 1.5, we obtain

$$|M_n| = (1 - o(1))|Z_n|. \tag{5}$$

7 Cleanup and completion

For each matched edge (x, y, P) , choose

$$a_P \equiv x \pmod{P}.$$

Since $|y - x| = 2P$, this also hits y . The residue is nonzero: if $P > n/5$ divided a residual target, then the target would be one of $P, 2P, 3P, 4P$, and these are nonresidual by the robust side-debt lemma.

Let

$$N = |A_S(n)| = (1 + o(1)) \frac{n}{\log n}.$$

By (5) and the definition of Z_n ,

$$|M_n| = \left((\beta - \frac{1}{5})\delta_S + o(1)\right) N.$$

The robust reservoir has size

$$|\mathcal{R}(n)| = \left(\frac{4}{5}\delta_S + o(1)\right) N.$$

Let $E_S(n) = o(N)$ be the exceptional residual-token count outside the main one-token set $A_S(n)$. After the pair stage, the unused robust-prime reservoir has size

$$U = |\mathcal{R}(n)| - |M_n|,$$

while the residual-token multiset still needing singleton cleanup has size

$$T_{\text{rem}} = |A_S(n)| - 2|M_n| + E_S(n).$$

Thus the exact injection condition is

$$U \geq T_{\text{rem}},$$

equivalently

$$|M_n| \geq |A_S(n)| + E_S(n) - |\mathcal{R}(n)|.$$

Using the asymptotics above and $E_S(n) = o(N)$, this follows from the strict inequality

$$(\beta - \frac{1}{5})\delta_S > 1 - \frac{4}{5}\delta_S,$$

which is exactly (1).

Assign each remaining residual token injectively to an unused robust prime P , choosing $a_P \equiv x \pmod{P}$. If an integer has two residual tokens, use two distinct unused robust primes. The robust side-debt lemma ensures that switching these primes creates no new unresolved debt. All unswitched primes remain in their zero classes, 2 remains in the odd class, 3 remains in the zero class, and the primes in S remain in the fixed nonzero classes. Thus every integer $m \leq n$ receives at least two hits.

8 References and status

The analytic input should be cited as the Green–Tao finite-complexity theorem for systems of affine-linear forms in primes, made unconditional using the Green–Tao nilsequence theorem and the Green–Tao–Ziegler inverse theorem for Gowers norms. The original linear-equations-in-primes framework is Green and Tao, *Linear equations in primes*, Ann. of Math. 171 (2010), 1753–1850.

The matching input is Jeff Kahn, *A linear programming perspective on the Frankl–Rödl–Pippenger theorem*, Random Structures and Algorithms 8 (1996), 149–157. Public metadata/previews appear to state the pair co-load parameter

$$a(t) = \max_{x \neq y} \sum_{A \ni x, y} t(A)$$

and the needed fractional-rounding conclusion. A final version should check the printed theorem statement directly.